

Carhart et al. (2002) estimate that the difference in risk-adjusted returns between live funds and nonsurviving funds is 3.7%. Malkiel and Saha (2005) report that the difference between the returns of live and defunct funds is 4.3%. Unfortunately for investors, mutual fund companies emphasize the returns only of existing funds. Careful academics always use databases like Carhart's, which include both surviving funds and dead funds; without the dead funds, mutual fund performance is highly overstated.

3.2. (UNDER-)PERFORMANCE

The short summary of the enormous mutual fund literature is that the average active mutual fund manager underperforms after fees but slightly beats (or at best equals) the market before costs.²⁶ This literature originally focused just on performance and was started by Treynor (1965), Sharpe (1966), and Jensen (1969). Michael Jensen, now an emeritus professor at Harvard Business School, wrote the article that gets most of the citations.²⁷ These papers found that active mutual funds do not beat the market, and these results have been echoed in many follow-up papers.

Russell Wermers' (2000) study summarizes the literature nicely. Over his sample, the S&P 500 returned 15.4% per year. Mutual funds beat the markets before fees, with a gross 16.9% return, but underperformed after fees, with a net return of 14.6%. This is an underperformance of 0.8%. Mutual funds with past high returns, however, could just have loaded up on risky stocks in rising markets. After adjusting for market and other factor risks (notably size, value-growth, and momentum factors; see chapter 7), Wermers finds a return in excess of risk factors (or *alpha*; see chapter 10) of 0.8% before fees. After fees, mutual funds have a return in excess of risk of -1.2%.

Eugene Fama and Kenneth French (2010), two giants of empirical asset pricing at the University of Chicago and Dartmouth College, respectively, estimate the alpha of equity mutual funds before fees is 0.2%, which is statistically equal to zero, and -1.1% after fees controlling for just the market factor.²⁸ Active mutual funds

the "reverse survivorship bias" is small: for a typical fund the Fama-French (1993) alpha is -0.44% per year while the alpha taking into account reverse survivorship bias is -0.41% per year.

²⁶ Other important studies not mentioned in this paragraph are Grinblatt and Titman (1989), Sharpe (1992), Malkiel (1995), Gruber (1996), and Fama and French (2010). See Ang, Goetzmann, and Schaefer (2011) for further literature review.

²⁷ Jensen is most famous for introducing the radical idea of paying corporate managers with options. His article Jensen and Murphy (1990) led to the profusion of option payouts in the 1990s—this practice went hand-in-hand with the Internet bubble of the 1990s. Treynor (1965), Sharpe (1966), and Jensen (1969) are also notable because they were the first applications of the CAPM. Treynor and Sharpe were two of the prominent developers of this theory (see chapter 6).

²⁸ The average bond mutual fund also does not add value. See Blake, Elton, and Gruber (1993) and Ferson, Henry, and Kisgen (2006). Institutional funds also underperform, as shown by Busse, Goya, and Wahal (2010).